23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No :D2

Team Members :

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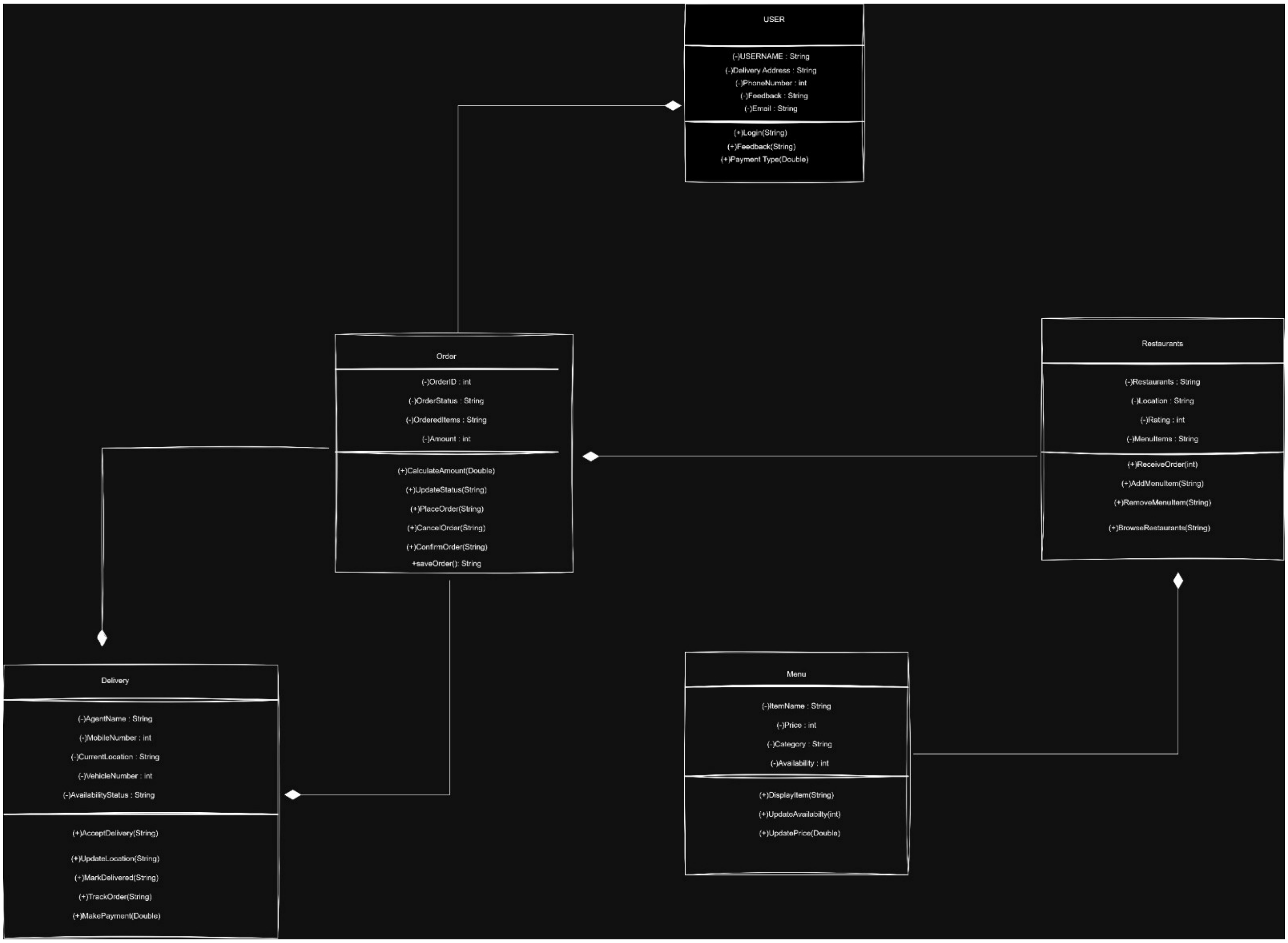
Problem Title : Online Food Delivery Management System

STRUCTURAL DESIGN

Class diagram (*draw class diagram that shows different classes and their relationships. Use UML notations*)


USER
(-)USERNAME : String
(-)Delivery Address : String
(-)PhoneNumber : int
(-)Feedback : String
(-)Email : String
(+)Login(String)
(+)Feedback(String)
(+)Payment Type(Double)
Order
(-)OrderID : int
(-)OrderStatus : String
(-)OrderedItems : String
(-)Amount : int
(+)CalculateAmount(Double)
(+)UpdateStatus(String)
(+)PlaceOrder(String)
(+)CancelOrder(String)
(+)ConfirmOrder(String)
+saveOrder(): String
Restaurants
(-)Restaurants : String
(-)Location : String
(-)Rating : int
(-)MenuItems : String
(+)ReceiveOrder(int)
(+)AddMenuItem(String)
(+)RemoveMenuItem(String)
(+)BrowseRestaurants(String)
Delivery
(-)AgentName : String
(-)MobileNumber : int
(-)CurrentLocation : String
(-)VehicleNumber : int
(-)AvailabilityStatus : String
(+)AcceptDelivery(String)
(+)UpdateLocation(String)
(+)MarkDelivered(String)
(+)TrackOrder(String)
(+)MakePayment(Double)
Menu
(-)ItemName : String
(-)Price : int
(-)Category : String
(-)Availability : int
(+)DisplayItem(String)
(+)UpdateAvailability(int)
(+)UpdatePrice(Double)

BEHAVIOURAL DESIGN

Use-case diagram

a. List of actors
   1. Customer
   2. Restaurant
   3. Admin
   4. Delivery Agent
b. List of use-cases:

Customer Use Case:

Register / Login — The customer creates a new account or logs into an existing one to access the platform.
Browse Restaurants — The customer views a list of available restaurants, filtered by location, cuisine, or ratings.
Browse Menu — The customer explores the menu of a selected restaurant, viewing items, descriptions, and prices.
Place Order — The customer selects items from the menu and submits an order to the restaurant.
Make Payment — The customer completes the transaction using available payment methods (card, wallet, UPI, etc.).

Restaurant Use Cases

6. **Register Restaurant** — A restaurant owner onboards their restaurant onto the platform with details like name, location, and cuisine type.

7. **Receive Order** — The restaurant gets notified of a new incoming order placed by a customer.

8. **Confirm Order** — The restaurant accepts or rejects the incoming order and begins preparation.

9 . **Add Menu Item** — The restaurant adds a new dish or product to their menu with name, price, and description.

Delivery Agent Use Cases:

10 . **Accept Delivery** — The delivery agent accepts an assigned delivery request for a prepared order.

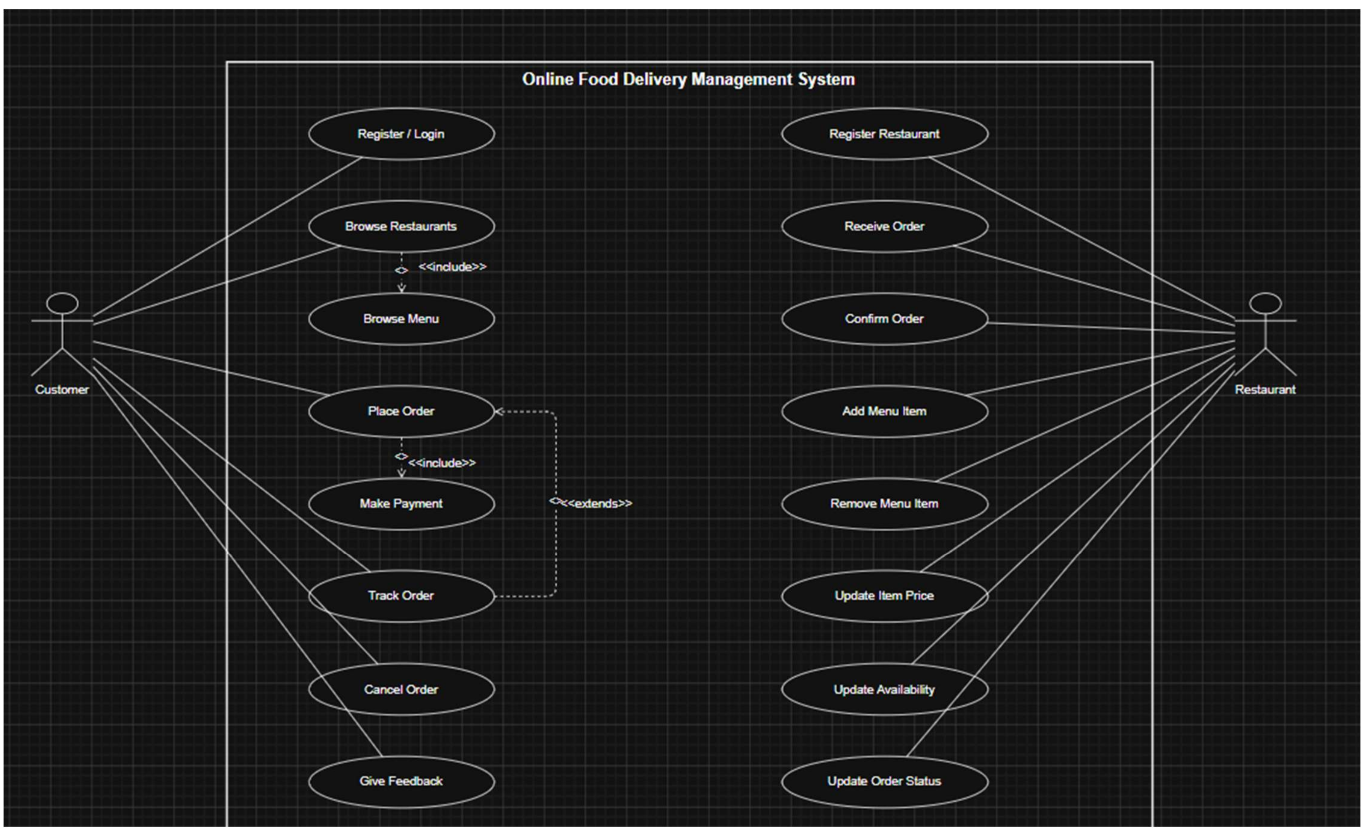
11. **Update Location / Mark Delivered** — The agent shares their live location during delivery and marks the order as delivered upon completion.

Admin Use Cases

12. **Manage Accounts & Orders** — The admin oversees all user accounts (customers, restaurants, agents) and monitors orders across the platform.

13. **Handle Disputes / Refunds** — The admin resolves complaints, mediates disputes between parties, and processes refund requests.

c. Use case diagram (use UML notations)



Sequence diagram (use UML notations)

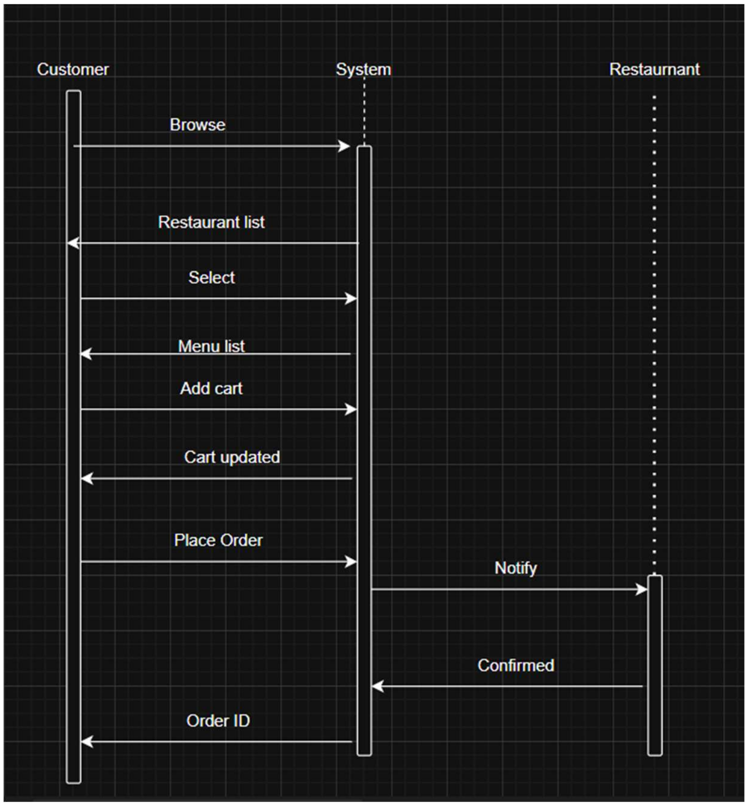
a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

Ans)

Browse and place Order:

Scernario: The User browses the restaurants shown by the system.After the user selects the restaurant, the system shows its meny, then the user adds items to cart and places the order.

Then the system gives a nofification to the restaurant and its confirms the order and the system gives an orderID to customer.

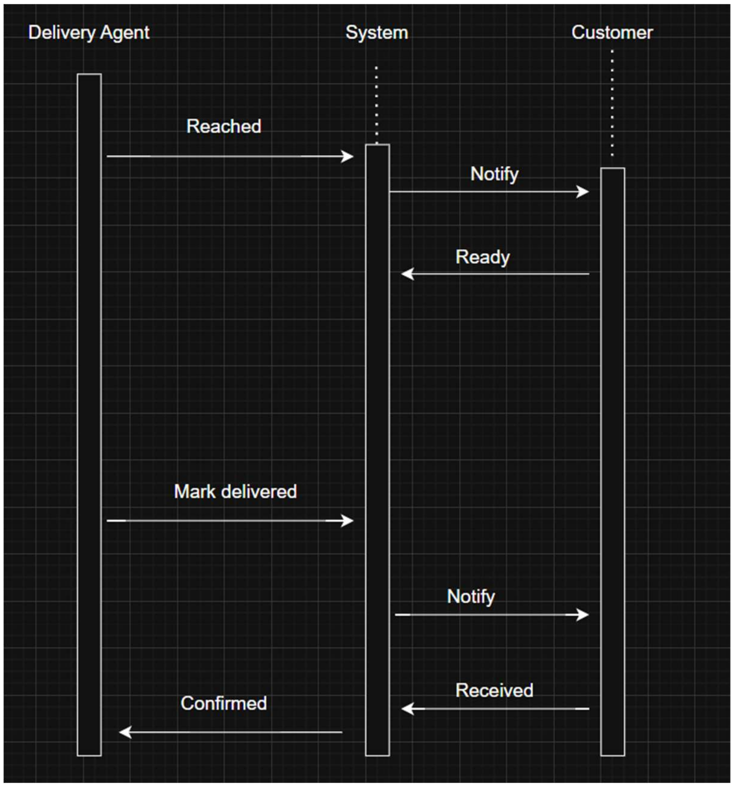


b. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

Ans)

Deliver order:
Scernario:
Once the delivery agent picks up the order from restaurant and reaches the users location,

They send a reached message to the system, then the system sends a notification to the user that the agent has arrived, the user now confirms that they are ready to received the order, the agent then marks the order as delivered, and the system notifies the use and the system confirms delivery finished to the agent.


Delivery Agent
System
Customer
Reached
Notify
Ready
Mark delivered
Notify
Received
Confirmed